Nicholas A Davies

Email: nicholasdaviees@gmail.com | **Phone:** (209) 417-8722 | **Active Security Clearance:** Secret | **LinkedIn** | **GitHub**

EDUCATION

California State University, Stanislaus **GPA: 4.0**

Bachelor of Science (B.S.) in Computer Science, Minor in Spanish *Expected Graduation: May 2027*

Honors/Awards: Honors Program, Dean's List, Warrior Excellence Scholarship, and NSF S-STEM Scholarship

SKILLS

Languages: C/C++, Python, Java, Bash

Technologies & Tools: Linux, Git, Claude Code, Docker, Kubernetes, CI/CD, GTest, GMock, TCP/IP Routing, ECMP, QoS

Other: Agile, Scrum, Jira, Confluence

EXPERIENCE

Keysight Technologies Software Engineer Intern *May 2026 - Present*

Keysight Technologies, Calabasas, CA

- Doubled Keysight's Fabric Emulator's per-port emulation speed from 400GE to 800GE by migrating to a next-generation ASIC, allowing it to support larger emulated data-center topologies at higher speeds.
- Unblocked the Fabric Emulator's transition to the new ASIC platform by refactoring the low-level ASIC C/C++ APIs for a newer SDK architecture, implementing TCP/IP routing functionality such as ARP, NDP, L2/L3 routing, ECMP, and QoS.
- Quadrupled port capacity by integrating a network-switch BSP and implementing port and SerDes lane mappings across Python, C++, and hardware configuration layers, while connecting software components with gRPC and Protocol Buffers.
- Restored reliable software delivery by reviving legacy CI/CD pipelines and automating build and deployment workflows using Bash, GitLab CI/CD, Docker, Kubernetes, and JFrog Artifactory.

Lockheed Martin Software Engineer Intern *June 2025 - August 2025*

Lockheed Martin MFC, Orlando, FL

- Improved reliability and accelerated delivery of mission-critical missile software by developing unit tests for C++ real-time embedded systems using GTest, CMake, GitLab, and CI/CD workflows.
- Diagnosed and resolved a persistent timekeeping defect in embedded C++ software, restoring accurate and reliable telemetry reporting for a critical internal system.
- Led the standardization of unit testing practices by authoring and delivering comprehensive documentation adopted across the JASSM software engineering team.
- Collaborated with multiple cross-functional Agile teams to deliver high-quality software through Jira-based task tracking, daily stand-ups, code reviews, continuous integration/continuous deployment workflows, and sprint planning.

Medical Big Data Analysis STEM-CRU Research Intern *September 2024 - May 2025*

California State University, Stanislaus

- Implemented a medical big data analysis application to identify disease–genome correlations using a 10 TB dataset under a 500 GB memory constraint, employing Apache Spark, GlusterFS, MongoDB, SparkR, and Shiny.
- Designed a secure user authentication and management system using Python, Linux, MongoDB, and Shiny.
- Performed statistical analysis and visualization using R, Python, Pandas, NumPy, Plotly, and cloud-based environments (Google Cloud, Azure ML).

Information Technology Support Specialist *August 2024 - Present*

Canal Veterinary Hospital, Turlock, CA

- Upgraded network infrastructure by deploying Ubiquiti routers and access points, expanding Wi-Fi coverage by 60%, optimizing firewall rules, and deploying an encrypted VPN to significantly strengthen data security.
- Maximized hospital workflow efficiency by migrating IT technologies from multiple legacy ecosystems into a unified AI-powered veterinary management platform, configuring and supporting 44+ hospital devices.

TECHNICAL PROJECTS & PROFESSIONAL DEVELOPMENT

ESP32 Dog Meal Tracker *November 2025*

- Built an embedded dog meal tracking system in C/C++ on an ESP32 microcontroller with a touchscreen interface.
- Engineered the TFT touchscreen GUI with custom bitmap pixel arrays and ESP32 GPIO input handling.
- Flashed and debugged firmware using the Arduino IDE, and utilized Git to manage development iterations.
- Implemented the embedded system with Apple HomeKit integration for real-time remote device control via iOS devices.

Spotify Music Listening Analyzer *February 2026 - Present*

- Designed a full-stack Spotify analytics platform using an HTML, CSS, and JavaScript frontend and a Python (Flask) backend API to process and analyze raw JSON streaming data into interactive user-facing insights and dashboards.
- Built a natural language query system powered by the Qwen2.5 LLM to translate user questions into structured analysis workflows, enabling conversational exploration of listening data.

Self-Hosted Infrastructure Platform *July 2020 - Present*

- Constructed a personal homelab architecture powered by Proxmox VE to manage Linux, Windows, and TrueNAS virtual machines, establishing self-hosted services in Docker containers via Docker Compose (YAML manifests).
- Advanced home network security and accessibility by defining VLANs and firewall rules, and utilizing Raspberry Pis as local DNS servers (Pi-hole), along with Tailscale VPN, Cloudflare tunneling, and NGINX reverse proxy.
- Automated system maintenance and monitoring with Bash scripts and cron jobs, reducing manual overhead by 95%.